

The role of instruction preference in analogy learning: Brain activity and motor performance

Tina van Duijn^{a,*}, Hamish Crocket^a, Rich S.W. Masters^{a,b}

^a Te Huataki Waiora Faculty of Health, Sport and Human Performance, University of Waikato, Hamilton, New Zealand

^b School of Public Health, Li Ka Shing Faculty of Medicine, University of Hong Kong, Hong Kong SAR, China

ARTICLE INFO

Keywords:

Analogy

Motor learning

Verbal

EEG alpha power

EEG coherence

Instruction preference

ABSTRACT

This study examined the role of verbal instruction preference when learning motor skills by analogy. During skill learning, analogies are a useful tool for providing knowledge about how to move. It has been argued that analogy instructions reduce reliance on verbal information processes during motor planning, compared to traditional forms of instruction (i.e., explicit rules about how to move). This may be reflected by reduced verbal activity in the brain, measured by EEG alpha power at the temporal region, as well as reduced verbal-motor cross-communication (EEG T7-Fz coherence) during the preparation phase of a movement. Preference for using verbal or visual instructions is likely to influence the efficacy of analogy instructions. This study investigated whether preference for verbal instructions was related to a) changes in performance and b) changes in verbal-cognitive information processing during performance of an adapted basketball task after instruction by analogy. Basketball novices with a high preference for verbal instructions ($n = 15$) showed significantly decreased activation of verbal brain regions when they used the analogy (high-alpha power), but their performance remained stable. Novices with a low preference for verbal instructions ($n = 13$) did not show a significant decrease in activation of verbal regions, and their performance deteriorated significantly after introduction of the analogy instruction. It is likely that both cognitive and performance changes after analogy instruction depend on personal aspects of information processing, such as verbal preference.

1. Introduction

Analogy learning requires the mapping of structural features from a familiar concept to a novel concept (Gentner, 1983). In sport, analogy learning has been shown to be especially useful for novices. To teach a topspin forehand shot in tennis, for example, a novice can be told to “move your hand as if it is traveling up the side of a mountain” (Poolton, Masters, & Maxwell, 2006, p. 679), which provides a heuristic (Todd & Gigerenzer, 2000) for how to move. Motor learning by analogy is characterised by stable motor performance both under pressure (Lam, Maxwell, & Masters, 2009a) and when a cognitive task, such as decision making, is performed concurrently with the motor task (Jie et al., 2016; Masters, Poolton, & Maxwell, 2008; Tse, Wong, & Masters, 2017).

Although Gentner (1983) devised a structure mapping model that convincingly explains the mechanism underlying analogy learning, we cannot fully explain the benefits (compared to conventional explicit instructions) in motor learning. It has been suggested that analogy

learning may lead to reduced verbal-cognitive control of movement (Liao & Masters, 2001), which makes information processing more effortless and efficient (Zhu, Poolton, Wilson, Hu, et al., 2011). In typical forms of learning, explicit instructions or rules often must be consciously implemented during practice, which can be neither effortless nor efficient. Even in cases when no instructions are given, novices may actively test hypotheses about how to move, thus accumulating self-instructions or rules (e.g., Masters, 1992; Masters & Maxwell, 2004). In any case, novice performers are likely to come to rely on declarative, explicit (verbalisable) knowledge when learning a movement, and tend to consciously control their performance (Fitts & Posner, 1967). Analogy learning may limit the amount of verbal information that novices access at an early stage of learning, leading to lower reliance on declarative information and reduced processing demands during motor planning (Bobrowicki, Macpherson, Coleman, Collins, & Sproule, 2015; Masters, 2000; Masters & Poolton, 2012). This may in turn free up information processing capacity for other cognitive processes (Berry

* Corresponding author. University of Waikato, Faculty of Health, Sport and Human Performance, Gate 4 Hillcrest Road, 3216, Hamilton, Waikato, New Zealand.
E-mail address: tinavanduijn@gmail.com (T. van Duijn).

& Broadbent, 1984)¹ and allow stable performance in dual task and pressure situations (Maxwell, Masters, & Eves, 2003). In most motor analogy experiments to date, fewer verbalisable rules were reported by the participants learning by analogy compared to explicit instructions or discovery learning (i.e., uninstructed; Poolton, Masters, & Maxwell, 2007). The notion that analogy instructions may lead to reduced reliance on verbal-declarative knowledge to control a movement is further supported by evidence of reduced cognitive load (measured by a probe reaction time paradigm; Lam et al., 2009a). Van Duijn, Hoskens, and Masters (2018) also found that analogy learning may lead to higher general economy of verbal-cognitive processes, as indicated by brain activity measurements. Psychomotor efficiency, which represents reduced cognitive effort required for relatively superior performance of a movement, has previously been linked to electro-cortical measurements (e.g., Cooke et al., 2014; Crews & Landers, 1993; Haufler, Spalding, Santa Maria, & Hatfield, 2000; Hillman, Apparies, Janelle, & Hatfield, 2000).

Collectively, research points to less effortful, verbal-analytic information processing when analogy instructions are used to teach a movement skill. If verbal information processing is altered or improved by using an analogy, individual differences in information processing are likely to have an effect on these changes. In particular, a person's preference for visual or verbal instructions may play a key role. The visual-verbal preference dimension is a stable personality tendency (Mayer & Massa, 2003) differentiating between verbalisers, who tend to prefer verbal instructions, and visualisers, who tend to prefer images (Riding & Watts, 1997).² Instruction preference may affect the modality in which movement-specific information is processed (Milz, Pascual-Marqui, Lehmann, & Faber, 2016). Research has shown that cognitive style³ is reflected in specific brain activity during execution of tasks (McKay, Fischler, & Dunn, 2002; Riding, Glass, Butler, & Pleydell-Pearce, 1997). Glass and Riding (1999) found that verbalisers exhibited higher activation of verbal processing areas during a verbal task compared to visualisers, as measured by EEG (electroencephalographic) analyses. Also fMRI studies have assessed visual-verbal preference in combination with BOLD⁴ activity during different tasks. Visual preference measured by the Visualizer-Verbalizer Questionnaire (VVQ, modified by Kirby, Moore, & Schofield, 1988) was shown to be positively correlated with BOLD activity at cortical areas associated with visual WM, while verbal preference was related to activation of areas associated with verbal WM during a colour knowledge retrieval task (Hsu, Kraemer, Oliver, Schlichting, & Thompson-Schill, 2011) and during visual and verbal tasks (Kraemer, Rosenberg, & Thompson-Schill, 2009).

Based on these findings, people with high verbal preference may also rely more on verbal processes to prepare and control their movements compared to people with a low verbal preference. Consequently, people with high verbal preference may therefore benefit most from introduction of an analogy, because it reduces verbal control processes (i.e., renders processing more efficient).

¹ Working memory (WM), the mental domain for short term storage and processing of information, has a limited capacity (Baddeley & Hitch, 1974) and thus limits the amount of cognitive information that can be processed concurrently.

² Concerns about the concept of learning styles and information preferences have been expressed (e.g., see Dekker et al., 2012); however, these doubts address the use of preference-matched instruction in school settings. The interest in preference in the present study is informed by the goal to test explanations for the benefits of analogy learning.

³ Cognitive style is a measure of preferred modality (visual or verbal) for information processes, based on Cognitive Styles Analysis by Riding and Cheema (1991). Although linked to similar aspects of personality, it is a separate construct from instruction preference.

⁴ BOLD is a form of functional brain mapping using blood oxygenation-level dependent contrast magnetic resonance imaging. It represents activity of the specified area at a high spatial resolution.

1.1. Measuring verbal-cognitive engagement in movement control

Measuring and quantifying verbal-cognitive information processes connected to motor control is an important step in determining the mechanism behind analogy learning. Indirect measures of verbal processing during motor preparation or execution include verbal protocols, dual task paradigms and probe reaction time tasks (Baddeley, 1966). However, verbal-cognitive contributions to motor planning processes can be measured more objectively using electroencephalography (EEG), a measure that assesses electrical potentials produced by brain cells through electrodes placed on the scalp. It allows analysis of the brain's activity in different areas of the cerebral cortex (see Klimesch, 1999, for a review). For reviews of research using psycho-physiological measures during sport and motor performance, see Cooke (2013) or Hatfield, Haufler, Hung and Spalding, 2004. Activation of brain regions measured during movement preparation reflects specific cognitive processes related to motor planning. The left temporal lobe (T7)⁵ contains verbal-analytic brain regions, such as Broca's area (i.e., syntactic processing and speech production) and Wernicke's area (auditory associations and speech comprehension; Clerget, Poncin, Fadiga, & Olivier, 2012). This region has been shown to be involved in verbal-analytic processing (Kaufer & Lewis, 1999; Sperry, 1974). Reduced alpha power (i.e., EEG power measured at 10–12Hz frequencies) in the left temporal region is associated with a high degree of conscious engagement in a motor task, and conversely increased alpha power in these areas suggests that verbal-analytical processes may be suppressed during performance (Hatfield, Landers, & Ray, 1984; Haufler et al., 2000; Kerick et al., 2001).⁶ Studies found higher alpha power and lower processing demands in successful compared to unsuccessful movements (e.g., Cooke et al., 2015; Crews & Landers, 1993), as well as in experts compared to novices (e.g., Haufler et al., 2000; Wolf et al., 2015), indicating that involvement of conscious control mechanisms during performance is gradually reduced as motor tasks are refined (Babiloni et al., 2010, 2008; Del Percio et al., 2010).

A direct indicator of verbal-cognitive contributions to motor planning is found in neural coactivation, or EEG coherence, a measure of functional communication between regions of the brain. EEG coherence between regions associated with motor planning (Fz, frontal midline) and verbal-analytic processing (T7) is thought to indicate the involvement of verbal processes in motor performance (Hatfield et al., 1984; Zhu, Poolton, Wilson, Maxwell, & Masters, 2011). Recent studies have found that experts display lower coherence between the T7 and Fz regions during movement execution compared to novices (Deeny, Hillman, Janelle, & Hatfield, 2003; Gallicchio, Cooke, & Ring, 2016). EEG high-alpha power and EEG coherence are therefore validated neural measures when investigating the effect analogy learning has on brain processes underlying performance.

1.2. The present study

We investigated whether preference for verbal instructions was related to changes in performance and in information processing (EEG high-alpha power and coherence at verbal and visual sites) immediately after a task-specific analogy was instructed. Basketball novices performed a block of 20 trials of a seated basketball task before and after an analogy instruction was provided. Participants' instruction preference was determined using the Verbalizer-Visualizer Questionnaire

⁵ In this paper, all labeling is translated to the nomenclature suggested in the American Electroencephalographic Society's "guidelines for standard electrode position nomenclature" (American Electroencephalographic Society, 1994; Chatrian et al., 1985; Jurcak, Tsuzuki, & Dan, 2007; Klem, Lueders, Jasper, & Elger, 1999). The terms T7 and T8 are used for left and right temporal lobes, while Fz is used for the frontal lobe site. Odd numbers indicate left hemisphere, even numbers right hemisphere locations.

⁶ Comparatively, the pre-motor area (Fz) is crucial in movement planning, while activity in the right temporal lobe (T8) has been shown to represent visuospatial mapping of movements (Wolf et al., 2015).

(VVQ, Kirby et al., 1988). Brain activity (EEG) was measured during motor preparation to examine whether changes in verbal or visual information processing were related to scores on the VVQ. Performance was expected to improve after introduction of the analogy, and more strongly so for people with a high verbal preference. Furthermore, a stronger increase in high-alpha power during movement preparation was expected for the verbal preference group.

2. Method

2.1. Participants

Initially, 74 participants (mean age = 19.35 years, SD = 2.76, 38 females and 36 males) with normal or corrected-to-normal vision and no movement impairment participated in the study. Participants were recruited from a sports science undergraduate class. Ethical permission for the study was received from the ethical committee of the institution. All participants provided informed consent. Participants with over 200 h of basketball experience ($n = 12$) were excluded from the analysis. Only participants who used incorrect shooting technique at pre-test ($n = 32$) were included, which reduced the sample by a further 16 participants. Correct or incorrect shooting technique at pre-test was judged by two independent raters using video footage (side-on perspective). Correct technique was defined as matching the movement described by the analogy instruction of “putting cookies into a cookie jar on a high shelf” (i.e., high elbow, wrist flick, and nondominant hand guiding the throw). Raters gave a score between 0 and 5 representing how closely the participants’ movements matched these three criteria. Scores of 4 or 5 during the pre-test were categorized as “correct shooting technique”, lower scores were categorized as “incorrect”. Cases in which the raters differed (6.75%) were discussed. If consensus was not achieved, the participant was excluded from analysis (after this process, ratings of both raters agreed to 100%). Additionally, extreme outliers (i.e., values deviating from the mean by more than 3 x the interquartile range) in T7 high-alpha power ($n = 2$) and T8 high-alpha power ($n = 2$) were excluded. Participants were allocated to low ($n = 13$) and high ($n = 15$) verbal preference groups using a median split approach. Five participants with scores on the median were excluded from the analysis. The final sample (7 male, 21 female participants) showed a mean age of 19.25 years (SD = 2.67).

2.2. Materials

For the basketball shooting task, a soft, slightly deflated beach volleyball (diameter = 23 cm) and a junior indoor basketball setup with backboard (height = 45 cm, width = 71 cm) and hoop (diameter = 39 cm) were used. Participants were seated on a chair (height = 45 cm) with horizontal distance from the rim equal to 315 cm. Height of the basket was adjusted to seated reaching height (at full arm extension) plus 30 cm (see Fig. 1). A low pitched tone and a high pitched tone, spaced 4 s apart, were played from a set of speakers to signal to the participant to start mentally preparing for the next shot (low pitch) and to execute the shot (high pitch). Performance measures were obtained by manual analysis of video footage from a Sony RX10pi camera focused on the target.

2.2.1. Electroencephalography

EEG equipment included a wireless EEG/tDCS transmitter, a set of 8 measuring and 2 reference electrodes, conducting gel and electrode contact stickers (Neuroelectrics Barcelona, SLU, ESP; see Schestatsky, Morales-Quezada, & Fregni, 2013, for details about the equipment). As this system is wireless, it allows the participant to move (to a certain extent). The system was connected to a desktop computer and analysed using Neurosurfer software by Neuroelectrics. EEG activity was recorded from 6 silver/silver chloride (Ag/AgCl) electrodes on the scalp positioned using different sized neoprene caps with pre-defined holes. Two reference electrodes were placed at the earlobe using a clip. Where the earlobe was too small to hold



Fig. 1. Experimental set-up.

the clip, two stickrodes were attached to the left mastoid. Caps and electrodes were adjusted carefully in line with the 10–20 system (Chatrjian, Lettich, & Nelson, 1985) and checked by two technicians before the start of the experiment. Measurement electrodes were positioned at the left and right temporal lobes (T7, T8), at the frontal lobe (F3, F4), central midline (Cz), pre-frontal lobes (Fp1, Fp2) of the scalp. Signals were collected at a sampling rate of 500 Hz with 24-bit resolution and 0–125 Hz bandwidth. Prior to commencing the task, an appropriate level of impedance was achieved by adjusting electrode positions, the participant's hair and the amount of electrolyte gel. While an upper limit of 15 kOhm was used for inclusion, impedance levels were generally below 5 kOhm. The pre-session criteria required participants to (a) wash their hair on the evening/morning before testing, and (b) not consume caffeine in the 2 h preceding testing.

2.2.2. Visual-verbal preference

Verbal-visual preference was assessed using an adapted version of the VVQ (Verbalizer-visualizer Questionnaire, adapted by Kirby et al., 1988). The original measure, developed by Richardson (1977), asked 15 yes/no questions about preference for visual or verbal instructions and cognitive style. Based on issues with unidimensionality, Kirby et al. (1988) developed an extended version to include a separate visual and verbal dimension (treated as separate scales). Additionally, a dream vividness dimension was extracted from the original scale and treated as a separate dimension. The adapted VVQ contains 30 items, 10 each for visual preference, verbal preference, and dream vividness. The latter subscale was not used in the present study. Out of the remaining 20 items, half were reversed for each subscale (i.e., a ‘yes’ answer would indicate low preference for that type of instruction). Kirby et al. (1988) reported acceptable reliability of the verbal preference scale ($\alpha = .70$) and of the visual preference scale ($\alpha = .59$). Principal components analysis revealed good construct validity for each scale. A high correlation was evident between verbal preference and verbal ability ($r = .321$, $p < .005$) and between visual preference and spatial visualization ($r = .270$, $p < .005$).

2.3. Procedure

2.3.1. General procedure

Participants completed the VVQ questionnaire online before the experiment (as part of a class assignment); however, they were unaware of any connection with the experiment. They provided informed consent and completed the rest of the experiment in the laboratory. They were fitted with the Neuroelectrics cap and the EEG electrodes were attached at the T7, T8, Fz, F3, F4, Cz, Fp1, and Fp2 locations. The motor task required participants to perform a modified, seated basketball-shooting task. This modified task was adopted to reduce the difficulty of

Table 1
Descriptive statistics for verbal and visual preference.

	Verbal preference		Visual preference	
	Mean	SD	Mean	SD
Overall	5.14	2.37	2.00	1.61
High verbal preference group	7.07	1.10	1.93	1.79
Low verbal preference group	2.92	1.12	2.08	1.44

the task and thus shorten the length of the adaptation process (shorter shooting distance and lower rim height compared with a regular free throw position and a different ball) and to allow collection of data in a controlled laboratory environment. Additionally, head movements were kept at a minimum by this adaptation of the task, which is a prerequisite for EEG measurement. Participants performed the task in a pre-test and a post-test of 20 trials each, during which EEG was collected. The task required participants to throw the ball as accurately as possible into the basket from where the chair was positioned. Participants were told to keep their back off the backrest of the chair in order to generate enough force. A signal consisting of two beeps, spaced 4s apart was used to indicate the start of preparation and shooting phase. Participants were required to prepare for each shot during the preparation phase (after hearing the first signal) and to shoot after they heard the second signal. They were also informed that they should not rush the execution of their shot after hearing the second signal.

2.3.2. Instruction

For the pre-test, participants were told to shoot the ball into the hoop, and were informed about the scoring method (see below). No instructions were given about how the movement should be performed. Prior to the post-test, participants were instructed with a validated analogy for the seated basketball shooting task (Lam et al., 2009a; Lam, Maxwell, & Masters, 2009b), which required them to “shoot as if you are trying to put cookies into a cookie jar on a high shelf”. No other instruction was provided.⁷

2.4. Dependent variables and data analysis

2.4.1. Dependent variables

Performance was measured on-line during the experiment, using a 6-point scale developed by Hardy and Parfitt (1991), according to which a score of 5 was awarded for shots that went through the hoop without touching the rim or backboard, 4 for those which touched the rim, 3 for those which hit the backboard before dropping through the hoop, 2 for those that touched the rim but did not drop through the hoop, 1 for those that touched the backboard and missed the hoop, and 0 for a complete miss. Doubtful or unsure calls were reinvestigated using video recordings collected at normal speed (30 fps) from a camera focused on the backboard. Hardy and Parfitt (1991) reported that the test-retest reliability of this scoring system over a 3-day interval and at different levels of physical fatigue (induced by running) was moderate ($r = .54$). Lam et al. (2009a, 2009b) used the same scoring method in a seated basketball task and reported high reliability (Cronbach's $\alpha = .94$). Mean shooting score in each block (i.e., maximum score 5 points) was calculated. Items for both subscales of the VVQ were reversed for negatively worded items. Item scores were summed

⁷ The proportion of participants who altered their technique during the pre-test (which may reflect hypothesis testing) was judged by two independent raters using video footage. Two participants in the low VVQ group, (i.e., 15.4%), and 0 participants in the high VVQ group, (i.e., 0%) altered their technique. A Two Proportion Chi-Square test revealed that the proportions were not significantly different between the two groups (Pearson Chi-Square (1,28) = 2.485, $p = .206$), therefore these data were not analysed any further.

separately for each subscale, leading to cumulative score ranges of 0–10 for verbal preference and visual preference.

2.4.2. EEG data reduction/processing

No baseline subtraction was performed. Raw EEG signals were bandpass filtered at 1–30Hz. Signals were then resampled at 256 Hz. Data from the pretest and post-test were processed as 4s segments prior to movement initiation (which was recorded on the EEG timetrace during the experiment using manual markers). Blinks and eye movements are characterized by high potentials (Boudet, Peyrodie, Gallois, & Vasseur, 2006). Eye artefacts were excluded by an extreme measures approach. Epochs (0.25s length) within each 4s segment that contained signals above $70 \mu V^2$ were discarded. An average of 251.24 (SD = 68.43) epochs per participant in the pre-test and 237.18 (SD = 77.19) epochs per participant in the post-test were retained. A Fast Fourier Transform with a Hamming window taper and 50% overlap with a resolution of 0.49 Hz, was applied to each 4s segment. EEG high-alpha (10–12Hz) power was calculated for the 4s segment preceding each trial (e.g., Deeny et al., 2003). Manual manipulation of the trigger has potential to compromise the validity of the power and coherence measures because of experimenter error, but considerable care was taken to avoid this.

2.4.3. Calculation of EEG power and coherence

The EEG analyses in this study focused on the high-alpha (10–12 Hz) band. Activity in this bandwidth indicates long- and medium-range cortico-cortical communication (Crews & Landers, 1993; Janelle et al., 2000; Nunez et al., 1999; Nunez & Srinivasan, 2006) and is representative of task-specific attentional processes (Smith, McEvoy, & Gevins, 1999; for a review see Klimesch, 1999).

Matlab scripts (MATLAB Release 2016b; The MathWorks, Inc., Natick, MA - US) were used to calculate EEG outputs. The processing and analysis steps described above were implemented with the EEGLAB toolbox (Delorme & Makeig, 2004). EEG coherence was calculated using a formula based on Equation (1), in which P_{xx} and P_{yy} represent the power spectral density of signals at location x and y, respectively, and P_{xy} represents the cross power spectral density of x and y (e.g., Zhu, Poulton, Wilson, Maxwell, et al., 2011). The measure was not z transformed. Coherence was analysed for T7-Fz and T8-Fz regional combinations separately, as these areas represent verbal-analytic (T7), visuospatial (T8) and motor planning (Fz) regions (Haufler et al., 2000; Kaufer & Lewis, 1999; Kerick et al., 2001).

$$C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_{xx}(f)P_{yy}(f)} \quad (1)$$

2.4.4. Statistical analysis

The assumptions of normal distribution, homogeneity of variance and sphericity were met. Performance scores, EEG high-alpha power and EEG coherence of high and low verbal preference groups were analysed using 2 (verbal preference: low, high) x 2 (test: pre-analogy, post-analogy) ANOVA's with repeated measures on the second factor (test). Follow-up t-tests were conducted to investigate changes within preference group if there was a significant interaction. Alpha value for statistical significance was set to .05. Since these tests were decided on a-priori, i.e., based on specific expectations associated with our rationale, a correction was not deemed necessary - see for example Rothman (1990).

3. Results

3.1. Verbal-visual preference

Table 1 shows the descriptive statistics for verbal and visual preference.

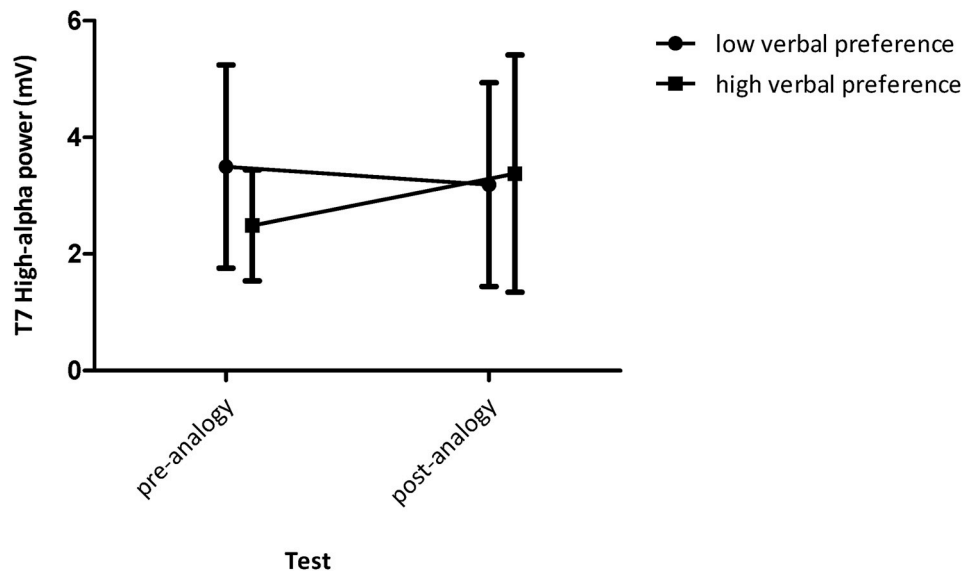


Fig. 2. High-alpha power at T7 (verbal processing area) in low and high verbal preference groups during the pre-analogy and post-analogy performance test.

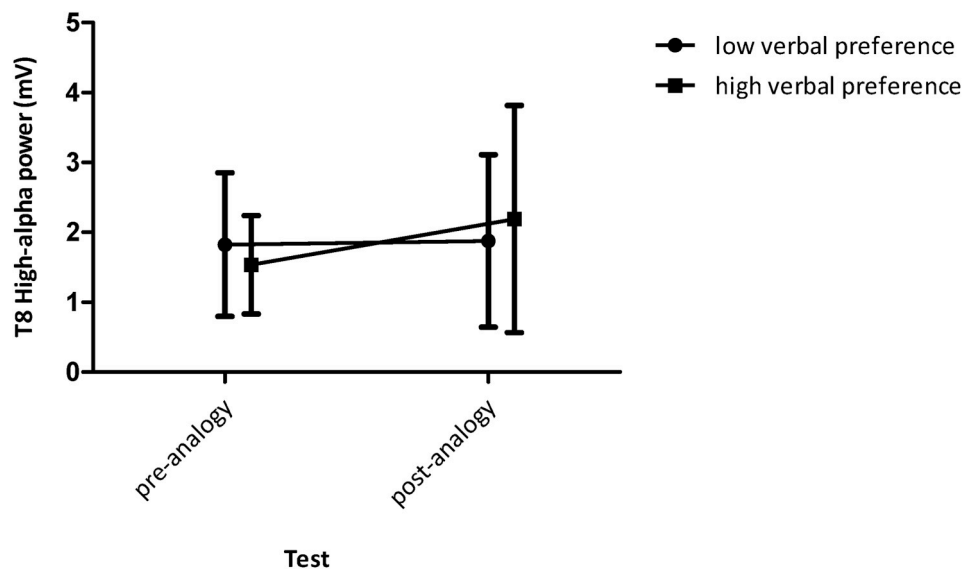


Fig. 3. High-alpha power at T8 (visuospatial processing area) in low and high verbal preference groups during the pre-analogy and post-analogy performance test.

3.2. EEG high-alpha power

Two separate 2 (verbal preference: low, high) $\times$ 2 (test: pre-analogy, post-analogy) ANOVA's with repeated measures on the second factor were conducted to investigate differences in high-alpha power at T7 and at T8. The analysis of T7 alpha power revealed no significant main effect of test ($F(1,26) = 2.44, p = .130$, partial $\eta^2 = .09$, observed power = .33) or verbal preference ($F(1) = .21, p = .654$, partial $\eta^2 = .01$, observed power = .07). A significant verbal preference $\times$ test interaction was evident ($F(1,26) = 5.15, p = .032$, partial $\eta^2 = .17$, observed power = .17). Follow-up t -tests showed that T7 high-alpha power did not change significantly for the low preference group ($t(12) = 1.02, p = .326$), but increased significantly in the high verbal preference group ($t(14) = -2.18, p = .047$, see Fig. 2). A t -test was also conducted to investigate differences between the groups at pre-test. No significant difference in high-alpha power at T7 was evident ($t(17.951) = 1.50, p = .162$, equal variances not assumed). The ANOVA for high-alpha power at T8 revealed no significant main effect of test ($F(1,26) = 3.73, p = .064$, partial $\eta^2 = .13$, observed power = .46) or of verbal preference ($F(1,26) = .003, p = .959$, partial $\eta^2 = .001$, observed power = .05) and no significant verbal preference $\times$ test

Table 2

Descriptive statistics for high-alpha coherence.

		Pre-test		Post-test	
		Mean	SD	Mean	SD
T7 -Fz	Low verbal preference	0.35	0.08	0.37	0.12
	High verbal preference	0.38	0.12	0.43	0.09
T8-Fz	Low verbal preference	0.29	0.10	0.31	0.07
	High verbal preference	0.32	0.14	0.37	0.12

interaction ($F(1,26) = 2.96, p = .097$, partial $\eta^2 = .10$, observed power = .38, see Fig. 3).

3.3. EEG high-alpha coherence

Two separate 2 (verbal preference: low, high) $\times$ 2 (test: pre-analogy, post-analogy) ANOVA's with repeated measures on the second factor were conducted to investigate differences in T7-Fz and T8-Fz coherence. The analysis of T7-Fz coherence revealed no significant main effect of test (F

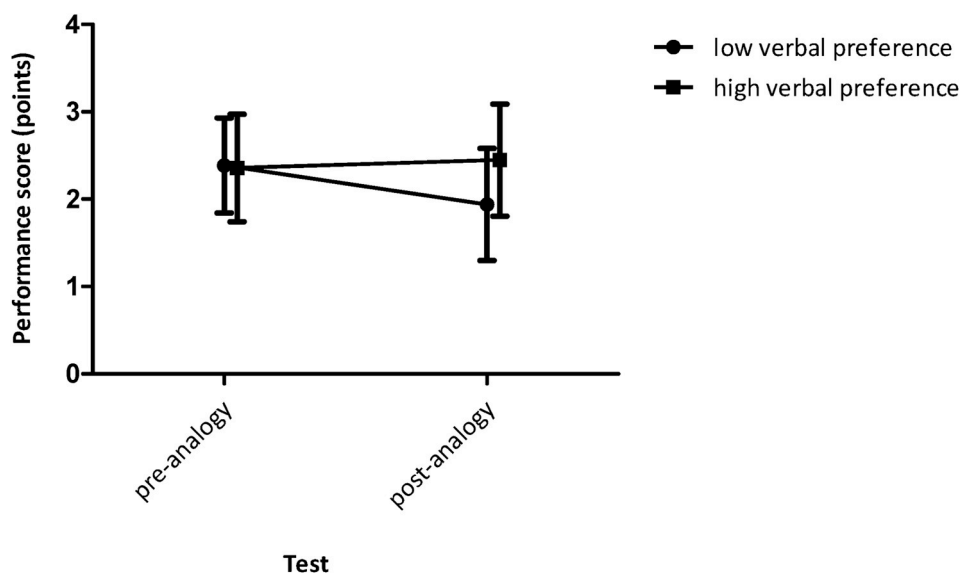


Fig. 4. Performance scores for low (dots) and high (squares) verbal preference groups in the pre-analogy and post-analogy test.

(1,26) = 3.27, $p = .082$, partial $\eta^2 = .11$, observed power = .41) or of verbal preference ($F(1,26) = 1.74$, $p = .199$, partial $\eta^2 = .06$, observed power = .25) and no significant verbal preference x test interaction ($F(1,26) = .94$, $p = .340$, partial $\eta^2 = .04$, observed power = .16). The analysis of T8-Fz coherence revealed a significant main effect of test ($F(1,26) = 4.71$, $p = .039$, partial $\eta^2 = .15$, observed power = .55), but no significant main effect of verbal preference ($F(1,26) = 1.23$, $p = .278$, partial $\eta^2 = .05$, observed power = .19) and no significant verbal preference x test interaction ($F(1,26) = 1.11$, $p = .301$, partial $\eta^2 = .04$, observed power = .17). Mean values and standard deviations of high-alpha coherence are shown for both groups in Table 2.

3.4. Performance

Mean performance scores for participants with low and high verbal preference are presented in Fig. 4. A 2 (verbal preference: low, high) x 2 (test: pre-analogy, post-analogy) ANOVA with repeated measures on the second factor was conducted to examine the performance scores. The analysis revealed no significant main effect of test ($F(1,26) = 3.41$, $p = .076$, partial $\eta^2 = .12$) or of verbal preference ($F(1) = 1.25$, $p = .275$, partial $\eta^2 = .05$). A significant verbal preference x test interaction was evident ($F(1,26) = 7.65$, $p = .010$, partial $\eta^2 = .23$, observed power = .76). Follow-up t-tests showed that performance decreased significantly in the low verbal preference group ($t(12) = 3.59$, $p = .004$), but no change was evident in the high preference group ($t(14) = -.62$, $p = .547$).

4. Discussion

The purpose of the present study was to investigate whether preference for verbal instructions was related to changes in performance and in information processing (EEG high-alpha power and coherence at verbal and visual sites) immediately after a task-specific analogy was presented. We will discuss the two dependent variables separately before synthesizing more general conclusions.

4.1. Motor performance

Motor performance remained stable from pre-to post-analogy in the high verbal preference group, while it deteriorated in the low verbal preference group. This indicates that performance after analogy instruction may be positively related to verbal preference. A potential explanation may be that people with high verbal preference generally have greater reliance on verbal-cognitive processes for planning and executing their movements,

which incurs higher processing demands.⁸ High cognitive demands may challenge novice motor performance by limiting the cognitive resources that are available for movement adjustments and the coordination of many degrees of freedom (Wulf & Weigelt, 1997). Indeed, by processing more movement-related information in the pre-analogy trials, high verbalizers may have performed under a higher cognitive load, although performance was still intact (as seen by nonsignificant difference between groups in the pre-analogy test). When implementing the analogy instruction, verbalizers may have experienced a greater reduction in verbal processing demands compared to low verbalizers, which would have facilitated movement coordination and led to performance benefits. Another cause for the difference in performance between the groups in the post-analogy test could lie in the differing ability of high and low verbalisers to process and apply verbal instructions. Although the analogy is a visually salient instruction that captures a complex concept within a simple comparison, it is instructed orally and is thus a verbal instruction (e.g., Poolton et al., 2006). Verbalisers may have simply benefited more from it as they were better able to use it or had fewer instances of misunderstanding.

Usually, the beneficial effect of analogy instructions on cognitive processes only becomes visible in dual task or pressure conditions, where an additional load on information processing resources is present. Performance of a regular task has uniformly been reported to be comparable to traditional (explicitly instructed) control groups (Bobrownicki et al., 2015; Tse et al., 2017). The results of the current study are somewhat at odds with these results, since we found that high and low verbalisers showed differences in their performance in a single-task test after analogy instruction. However, this study differs from previous studies as we investigated immediate effects after introduction of analogy (without practice). The shortest periods of practice after introduction of the analogy in the existing literature consisted of 40 trials (Bobrownicki et al., 2015) or around 80 trials (Koedijker et al., 2011; Vine, Moore, Cooke, Ring, & Wilson, 2013).⁹ Since the present experiment did not include a control group that received a different set of instructions (either explicit instructions or no instructions), we cannot compare analogy learning to explicit

⁸ The amount of information that was acquired by participants was not analysed in this study.

⁹ In a walking study by Jie et al. (2016), as few as 8 practice trials were included; however, one trial of a 10 m walk included around 10 repetitions of the step pattern, resulting in an estimated total of 80 practice trials of the step pattern. Furthermore, patients were asked to also train at home (Jie et al., 2016).

learning results after 20 practice trials.

The fact that overall performance did not improve from pre- to post-analogy tests may be a consequence of the low number of practice trials in this study. The demands of processing new instructions can lead to an initial decrease in motor performance, presumably because they reduce the capacity available for the adjustment of movements during learning (Hodges & Lee, 1999; Wulf & Weigelt, 1997). The notion that effective use of an analogy may require some time and cognitive effort is supported by a study by Masters and Liao (2003), which reported a decrease in performance immediately following instruction by analogy.

4.2. Verbal-cognitive processing and verbal-motor coherence

Participants with a high preference for verbal instructions showed an increase in high-alpha power at the left temporal lobe after they were instructed with an analogy, while participants with a low verbal preference did not show any changes in cortical activity. An increase in high-alpha power over the left temporal lobe has been linked to a decrease in cortical activation in this region, suggesting that fewer verbal processing resources are used (Cooke, 2013; Hatfield et al., 1984). The present finding may thus indicate reduced occurrence of verbal processes during motor planning; however, increased high-alpha power was only found in people who preferred verbal instructions. This finding supports the notion that high verbalizers may have, by relying more on explicit, verbal control processes, incurred higher processing demands during motor planning (Maxwell et al., 2003). Upon introduction of the analogy instruction, verbalizers may therefore have exhibited a greater drop in verbal-cognitive processing compared to low verbalizers. However, EEG alpha power measured at pretest did not differ significantly between the groups. As an alternative explanation, we propose that higher left-temporal high-alpha power (i.e., lower activation) in the verbal preference group could be related to more efficient processing of the analogy instruction itself. High-alpha power reflects verbal-cognitive processing regarding a planned movement, but it is also representative of other verbal processes that may be independent of motor planning. Verbal preference is potentially related to verbal ability (e.g., Kirby, et al., 1988; but also see Alesandrini, 1981 for a contradictory finding), so verbalisers may be better at making abstract connections between the source (i.e., cookie jar) and the target of the analogy (i.e., movement), which is necessary for successful processing of analogies (Gentner, 1983; Sternberg, 1977). Better ability to process the analogy may thus explain why high verbalisers showed brain activity patterns reflective of more efficient verbal processing after analogy instruction. This explanation also fits with the performance data reported previously, as motor performance remained stable in the high verbal preference group, while it deteriorated in the low verbal preference group.

No difference between verbal preference groups was found in T7-Fz coherence, which indicates that neither group's verbal-motor cross-communication became more efficient. This is at odds with previous findings by Zhu and colleagues, which showed that implicit motor learning leads to a decrease in T7-Fz coherence (Zhu, Poolton, Wilson, Hu, et al., 2011; Zhu, Poolton, Wilson, Maxwell, et al., 2011). The only previous study of analogy learning that involved cortical activity measures found that high-alpha power at T7, but not T7-Fz coherence was affected by analogy (van Duijn et al., 2018). The authors argued that using an analogy may increase efficiency of verbal-cognitive processes, but verbal-motor cross-communication may remain the same. The mechanism responsible for the benefits connected to analogy in motor learning may be tied to verbal-cognitive processes, rather than verbal-motor cross-communication, and it might thus differ from the mechanism underlying other forms of implicit motor learning.

Hitherto, a potential explanation for analogy learning was suggested by Liao and Masters (2001) to be a shift from verbal to visual processing of information, based on the notion that analogies, being visual in nature, might be more easily processed in the visual subdomain of

working memory (Baddeley, 2012). Neither high-alpha power at T8 nor T8-Fz coherence showed any group by test interactions or main effects. This indicates that rather than shifting from verbal to visual processing (which would have produced an opposite effect in visual areas), information processing may have been rendered more efficient while remaining in the verbal domain of working memory. Further research is needed to compare the different suggested mechanisms directly.

4.3. Relative efficiency of information processing

When looking at cognitive and performance parameters in combination, differences between the instruction groups in psychomotor efficiency are evident. Psychomotor efficiency refers to a decrease in cognitive effort required for performance of a movement, concurrent with an increase in performance during development of expertise and has been linked to electro-cortical measurements such as higher alpha power (Cooke et al., 2014; Haufler et al., 2000; Hillman et al., 2000) or reduced cortico-cortical coherence (Hatfield & Hillman, 2001; Zhu, Poolton, Wilson, Maxwell, et al., 2011). More precisely, psychomotor efficiency may reflect either higher performance with reduced or stable neural effort, or maintenance of performance using less neural effort. It may therefore also indicate lower reliance on explicit, verbal information during preparation or performance of a movement (for a review see Hatfield, Haufler, Hung, & Spalding, 2004). In the present study, high verbalisers did not show a change in performance but exhibited decreased activation in the verbal brain region, which indicates increased efficiency of verbal-cognitive processes. Verbalisers thus became relatively more efficient in their verbal-cognitive processes. In contrast, the low verbalisers deteriorated in performance, without showing any change in brain activity, which equals a relative decrease in efficiency.

This finding supports one of the suggested mechanisms that may underpin analogy learning: the chunking hypothesis of analogy learning (Masters & Liao, 2003). Chunking is a process of grouping a higher number of small chunks into a smaller number of larger chunks by organizing patterns during learning (Newell & Rosenbloom, 1980). Masters and Liao (2003) suggested that analogy learning may be a way to chunk movement-specific information more effectively than in the long learning process, as the analogy provides “pre-chunked” information that is already on a higher organizational level. The finding that verbalisers showed a greater increase in psychomotor efficiency may be explained by a higher number of rules being reduced into a single, information-rich chunk (the analogy). Furthermore, the high verbal preference group might have acquired a more in-depth understanding of the movement components by accumulating more information. A more complete understanding of the movement may speed up chunking, since only items that are understood and that fit within a common higher-order structure can be chunked together (Baddeley, Thomson, & Buchanan, 1975).

5. Practical implications

These results also have practical implications for coaching. Individual characteristics related to preference, ability and personality may have to be taken into account when designing learning interventions – especially where these involve analogy instruction. A deeper understanding of the mechanisms underlying analogy learning may therefore also advance the improvement of skill instruction in various types of motor skills, including ontogenetic (complex, learned) skills such as tool manipulation, sports skills, etc. and phylogenetic (naturally evolving) skills such as walking. This may have impacts on clinical therapy, physiotherapy, surgery, aviation, and many other fields.

6. Limitations

Although the measure of visual-verbal preference used in this study, an adapted version of the Verbalizer-Visualizer Questionnaire

(Richardson, 1977), has been validated (Kirby et al., 1988) and used in previous studies (Hsu et al., 2011; Kollöffel, 2012; Kraemer et al., 2009; Milz, Faber, et al., 2016), it is not a validated instrument to measure preference in the sport context. The importance of developing setting-specific measures of learning preference has been stressed by Coker (2000), who showed significant intra-individual differences when a learning preference questionnaire was used in different settings, i.e. in a classroom setting vs. in a clinical sports setting. The importance of developing a validated tool applied to motor learning preferences has been mentioned before (Fuelscher, Ball, & MacMahon, 2012). The measure that is most widely used in the sports field, the VARK (Visual, Auditory, Reading/writing, Kinaesthetic, Dunn, 2009), has barely acceptable validity (Leite, Svinicki, & Shi, 2010). Although an athlete version of the VARK has been developed, the adapted questionnaire has not been validated and questionnaire items are not related to motor instructions (Fuelscher et al., 2012), which is why we did not choose to use this measure in the present study.

A further limitation to the study is our focus on experimental control, which limited the external validity of the task. As the results of the study have implications for sports practice and instruction, future studies (e.g., long-term, controlled learning studies in the field) should attempt to investigate whether the same effect persists in real sports skills. Additionally, the sample size in the present study was limited, partly due to the strict selection criteria. A small sample size may reduce the likelihood of finding significant results, however, significant effects were reported in the paper. It is therefore likely that replicating the study with a larger sample might corroborate and strengthen our findings. It is important that future research attempts to replicate the present findings and advance the current knowledge in this area of research.

7. Conclusion

The results of the present experiment indicate that motor performance changes as well as changes in cognitive efficiency after analogy instruction may be related to verbal preference. Analogy instruction may increase efficiency of verbal processes in high verbalisers, but not in low verbalisers. This generally supports the hypothesis that analogy reduces verbal processing demands by reducing the amount of verbal information being processed, providing some support for a chunking hypothesis of analogy instruction.

Declaration of competing interest

The authors have no conflicts of interest to declare.

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